

BRAD-CORE OS V5 UNIFIED: FORMAL SAFETY CASE & STATUTORY ASSESSMENT

Compliance Demonstration for DoD Directive 3000.09 (Autonomy in Weapon Systems) via G003 HITL-3 Fuse

Document ID:	BC-OS-CASE-DOD3000-001 Rev A
Statutory Target:	DoD Directive 3000.09 (§ 3 • § 4 Guidelines for Autonomous and Semi-Autonomous Weapon Systems)
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Hardware Domain:	SOSA / OpenVPX 3U/6U • G003 Tri-Seal Hardware Fuse • Emmy Clause Non-Bypassable Invariant

1. EXECUTIVE STATUTORY MAPPING

Department of Defense Directive 3000.09 mandates that autonomous and semi-autonomous systems must be designed to allow commanders and operators to exercise appropriate levels of human judgment over the use of force, guarantee predictable operational behavior under realistic contested conditions, and prevent unintended engagements caused by software drift, communications loss, or algorithmic runaway.

Conventional autonomy architectures attempt compliance via application-level software overlays, which are vulnerable to process deadlocks, queue starvation, and remote tampering. **Brad-Core OS v5 Unified** satisfies DoDD 3000.09 requirements by rooting human authority and fail-closed safety directly into silicon hardware via the **G003 HITL-3 Safety Fuse** and the **Emmy Clause**.

2. CLAIMS, ARGUMENTS, AND EVIDENCE (GOAL STRUCTURING NOTATION ALIGNMENT)

Claim 1: The system precludes unintended kinetic actuation or autonomous runaway.

- Argument:** No machine-learning or probabilistic path planner (P-Core) holds hardware write permissions to kinetic control surfaces or firing solenoids. Actuation authority requires continuous cycle-by-cycle proof mediated by the Deterministic Master (D-Core).
- Evidence:** Compile-time memory layout enforces `malloc = 0`. Sheriff Engine asserts sub-86 μs MPU page protection (`PROT_NONE`) over physical pins if invariant checks fail or cycle time exceeds `$T_{ ext{WCET}} \le 950 \mu ext{s}`.

Claim 2: Mode B HITL-3 satisfies human supervisory requirements without introducing manual bypass risks.

- Argument:** Mode B is not a steering override; it is an emergency latency fuse designed to preserve deterministic timing during extreme bus saturation. Human presence cannot be weaponized to suppress a machine safety fault.
- Evidence:** Section 9.2 of Hardware SDD BC-OS-SDD-HW-001 Rev A enforces the invariant `VALIDATE ^ CLAMP ^ HOLDORRAISE`. Effector firing is architecturally impossible (`ACT = 0`). Humans are not a compare channel.

“DoDD 3000.09 requires human judgment to prevent catastrophic autonomous failures. Mode B provides that judgment by clamping actuators and escalating safety states, while the Emmy Clause guarantees no human can vote to ignore a detected physical fault.”

+-----+ DOD DIRECTIVE 3000.09 HARDWARE COMPLIANCE MAPPING +-----+	
DoDD 3000.09 Mandate	Brad-Core OS Silicon Realization
Appropriate Human Judgment	Mode B HITL-3 Tri-Seal Quorum (3 Hardware Inputs)
Fail-Safe Under Loss of Comms	Automatic Return to H0 Safe State in <100 μs
Predictable Timing Bounds	1 kHz Anneliese Engine (T_WCET ≤ 950 μs, ≥ 72% Margin)
Anti-Tamper & Cyber Defense	Amy Protocol 3-Stage Cascade + <10 μs Zeroize
Constitutional Command Link	State H_NCA Presidential Gold-Code Asymmetric Token
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3. STATUTORY REQUIREMENT VERIFICATION MATRIX

Directive Clause	Statutory Requirement	Brad-Core OS Architectural Feature	Verification & Test Artifact
DoDD 3000.09 § 3.a	Systems shall be designed to allow commanders and operators to exercise appropriate human judgment over the use of force.	G003 HITL-3 Fuse (Mode B): Actuation bus registers remain clamped in PROT_NONE. System only executes under active human consensus.	Hardware pin-clamp continuity test; 72/72 red-team vector clearance.
DoDD 3000.09 § 3.b	System must function predictably under adversarial counter-measures, spoofing, and contested environments.	Amy Protocol v2/v3 Unified: Dedicated SOSA 3U/6U blade with supercapacitor rail; raw IMU cross-auditing against GPS spoofing.	Electronic Warfare injection test plate; 120-second host-blackout power continuity test.
DoDD 3000.09 § 4.a(1)	Safety, anti-tamper, and cyber-security controls to prevent unintended engagements.	The Emmy Clause: $\forall s \in \text{Signals}, \text{Status}(s) \neq \text{IGNORED} \wedge \text{Status}(s) \neq \text{DISMISSED}$. Zero-trace key dump in <10 μs on breach.	34 Constitutional Red-Team Vectors; Physical chassis tamper trigger blast test.
DoDD 3000.09 § 4.a(2)	System must fail safe upon communication loss, degradation, or unexpected hardware state.	Worst-Case Inversion & Sunset Timer: Default initialization to H0; 120-second silicon sunset timer; single-seal loss defuses Mode B in <10 μs .	Tri-seal withdrawal oscilloscope capture; monotonic frame timeout boundary logs.

4. OPERATIONAL ASSURANCE: THE FOUR-MODE DEFENSE ENVELOPE

The integration of DoDD 3000.09 into Brad-Core OS establishes a complete operational continuum:

- Mode A (Nominal Autonomy):** Uncrewed navigation and telemetry auditing within strict mathematical corridors under the Anneliese Engine pulse.
- Mode B (Emergency Latency Fuse / G003):** Human operator interlock under bus saturation. Reduces inter-core latency while strictly clamping effectors (HOLDORRAISE).
- Mode C (Amy Sovereign Containment):** Out-of-band silicon shutdown if software models experience containment breach or adversarial hijacking.
- State H_NCA (Apex Command):** Formal cryptographic pipeline ensuring the Commander-in-Chief retains lawful, non-delegable command authority.

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